Drug Category Prediction Based on Target Genes (Case Study: Human Brain)

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March 1, 2026

Abstract

This project investigates whether drug target genes and their brain cell-type-specific expression patterns can provide sufficient molecular evidence to support therapeutic category predictions. Using the Anatomical Therapeutic Chemical (ATC) classification as a reference, we propose a two-stage pipeline that first evaluates central nervous system (CNS) relevance and subsequently predicts high-level drug categories based on brain cell-type gene-expression features. Rather than claiming clinical correctness, the system is designed as an exploratory, hypothesis-generating decision-support tool. The brain serves as a case study to demonstrate how cell-type-restricted molecular evidence can support, question, or qualify predicted drug categories.

1 Introduction

A gene expression signature (GES) is a set of differentially expressed genes that reflects the transcriptional changes between a stimulated and a normal cellular state [QSF+20]. GES analysis has been increasingly applied in drug research and repositioning studies.

Qi et al.[QSF+20] evaluated the performance of GES-guided drug–disease association methods and showed that comparing gene expression profiles can help identify mechanistic similarities between drugs and therapeutic relationships between drugs and diseases. In particular, if the gene expression pattern induced by a drug is inversely correlated with that of a disease, the drug may have therapeutic potential. They also reported that integrating machine learning approaches can improve the accuracy of compound indication prediction.

Motivated by these findings, we use gene expression signatures associated with drugs to analyze similarity between drugs and predict their therapeutic categories. Since drug action is influenced by tissue-specific expression patterns, we focus on brain-related gene expression data to improve prediction performance for central nervous system (CNS) applications.

However, restricting the analysis to brain tissue may introduce bias. Therefore, we first classify drug target genes based on their Anatomical Therapeutic Chemical (ATC) category and then distinguish between CNS-related and non-CNS-related gene expression. This dual strategy helps reduce tissue bias and improve prediction reliability.

Our approach enables the prediction of the top therapeutic categories for a given combination of target genes, which may also support drug repositioning[LCPea06]. To facilitate practical use, we designed an interface that allows researchers to input target genes and obtain category predictions, probability scores, CNS relevance scores, and an explanation of the results.

2 Problem Statement

Drugs are commonly categorized according to their therapeutic function, such as antibiotics, anticoagulants, and analgesics. Previous studies suggest that therapeutic function is closely related to the genes targeted by a drug and the tissues or cell types in which these genes are expressed. In particular, gene expression within specific brain cell types plays a defining role in determining the effects of drugs acting on the central nervous system (CNS).

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However, therapeutic classifications such as ATC codes are defined primarily by clinical usage rather than cell-type-specific molecular evidence. This raises an important question: can brain cell-type-specific gene expression data help distinguish CNS-active drugs from non-CNS drugs and support therapeutic category prediction at an early stage of drug development?

The central problem addressed in this project is to operationalize this idea by:

- Predicting the therapeutic category of a new drug based on its target genes and their expression patterns across brain cell types.
- Explicitly distinguishing CNS-active drugs from non-CNS drugs, since the molecular relevance of brain cell-type expression is expected to be stronger for drugs acting on the nervous system.
- Investigating cross-category effects of drugs that may belong to multiple therapeutic classes, such as acetaminophen, which exhibits both analgesic and antipyretic effects.

By leveraging brain cell-type-specific gene expression data, this project aims to identify and track drug categories based on molecular signatures prior to experimental testing. The focus on CNS classification provides a biologically grounded validation step, ensuring that the use of brain-specific data aligns with clinically defined nervous system drugs.

3 Data Sources

3.1 DrugBank and DRUGseqr

Drug target information and ATC classifications are obtained from the DrugBank with and DRUGseqr database and the combine output file contains 28,393 drug entries. For each drug, target genes, enzymes, and transporters are extracted from the DrugBank XML. Only drugs with sufficient metadata are retained for model training. DRUGseqr contains drug-induced gene expression signatures from experimental perturbation data.

3.2 Brain Gene Expression Data [Hum24]

Single-nucleus RNA sequencing data of the human brain is used to construct a gene-by-cluster expression matrix. Expression values are aggregated per gene across defined brain cell clusters, forming the biological evidence basis of the project.After pre-processing, this dataset contains 665,720 gene-cluster pairs and 19,574 unique genes and 34 unique cluster types.

4 Methodology

The category prediction pipeline is designed as a Neuro-Symbolic system, where biological entities (genes, drug targets, therapeutic labels) are represented as symbolic components, and transcriptomic datasets together with numerical modeling constitute the statistical layer. This hybrid design enables structured biological knowledge to be integrated with data-driven learning.

To formally design and validate the architecture, we used Tool4Boxology [1], a framework developed at TIB for constructing and analyzing neuro-symbolic systems. The tool supports systematic modeling from scratch and provides structural validation of the interaction between symbolic and neural components. Based on the boxology diagram, the system is organized into six main tasks:

Task 1: Analyze single-cell RNA sequencing (scRNA-seq) data to characterize gene expression patterns across brain cell cluster types.

Task 2: Integrate DrugBank and DRUGseqr.gmt resources to extract drug–gene interaction information.

Task 3: Extract therapeutic classifications from DrugBank using ATC codes as symbolic labels.

Task 4: Generate feature embeddings by combining brain cell-type expression profiles, drug target genes, and gene expression signatures.

Task 5: Multi-Stage Drug Classification — CNS Activity and Therapeutic Category Prediction.

[1] Tool4Boxology, TIB – Leibniz Information Center for Science and Technology. Available at: https://tool4boxology.service.tib.eu/

Task 6: Learn predictive patterns to infer drug therapeutic categories based on gene effects and embedding representations.

These steps define the neuro-symbolic workflow: symbolic knowledge (ATC labels, drug–gene relations) is fused with neural embeddings derived from transcriptomic data, and the resulting representations are used for supervised learning.

At the application level, a user interface is connected to the trained backend model. Researchers can input the target genes of a new experimental drug, and the system generates top-k therapeutic category predictions with associated probabilities and evaluation metrics.

To improve interpretability, an explanation module (agent) summarizes and interprets the evaluation metrics, providing users with a clear justification of the predicted categories and CNS relevance scores. This final layer enhances transparency and supports informed decision-making in experimental settings.

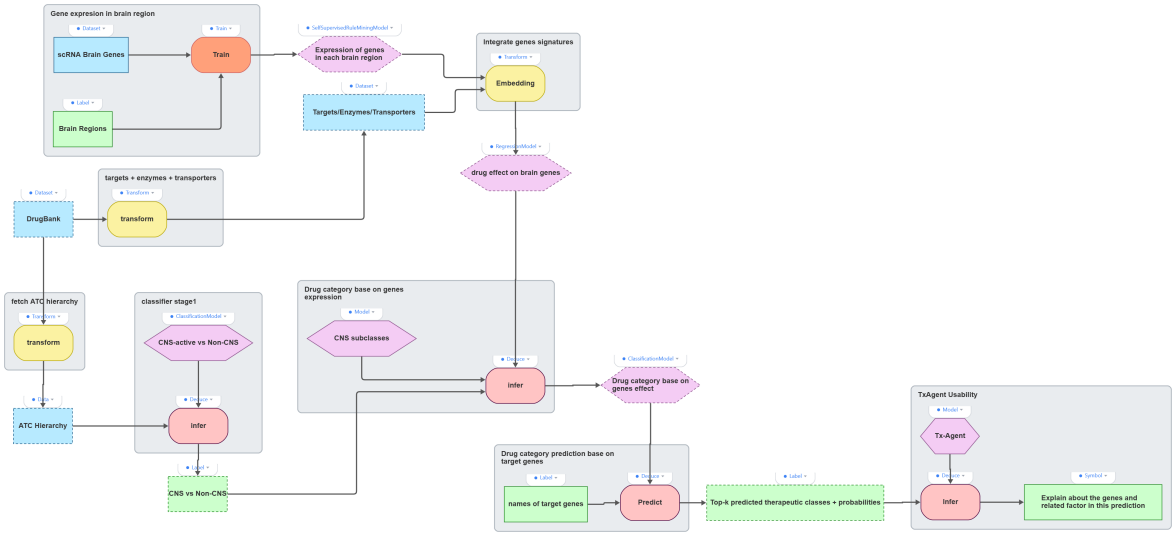


Figure 1: Boxology diagram illustrating the implementation of the proposed multi-stage learning and reasoning framework within a neuro-symbolic architecture.

4.1 Gene Expression Across Brain Cell Types

Task 1 is shown in Figure 2. The heatmap displays how genes are expressed differently across brain cell types. Some genes, such as NRXN1/3, DLG2, CNTNAP2, and SYT1, are more active in brain (neuronal) cells and less active in other cell types like fibroblasts and leukocytes. This shows that brain cells have specific gene activity patterns, which supports using brain data in the prediction model.

4.2 Drug–Gene Interaction Information

Task 2 integrates DrugBank and DRUGseqr.gmt to extract drug–gene interactions. The variance analysis shows that certain genes vary strongly across drugs, indicating their potential importance as discriminative features for category prediction [Figure 3]. Additionally, the distribution of gene types reveals that most interactions correspond to primary drug targets, confirming the biological relevance of the extracted dataset.

4.3 ATC Hierarchical Label Construction

Task 3 extracts ATC classifications from DrugBank while preserving their hierarchical structure. The number of unique categories increases from Level 1 to Level 3 (Figure 4a), reflecting increasing specificity (e.g., Nervous system → Psychoanaleptics → Antidepressants).

The Level 1 distribution 4b shows representation across multiple therapeutic domains. Additionally, the histogram 4c of ATC codes per drug indicates that while most drugs have a single ATC code, some belong to multiple categories, confirming the multi-label nature of therapeutic classification.

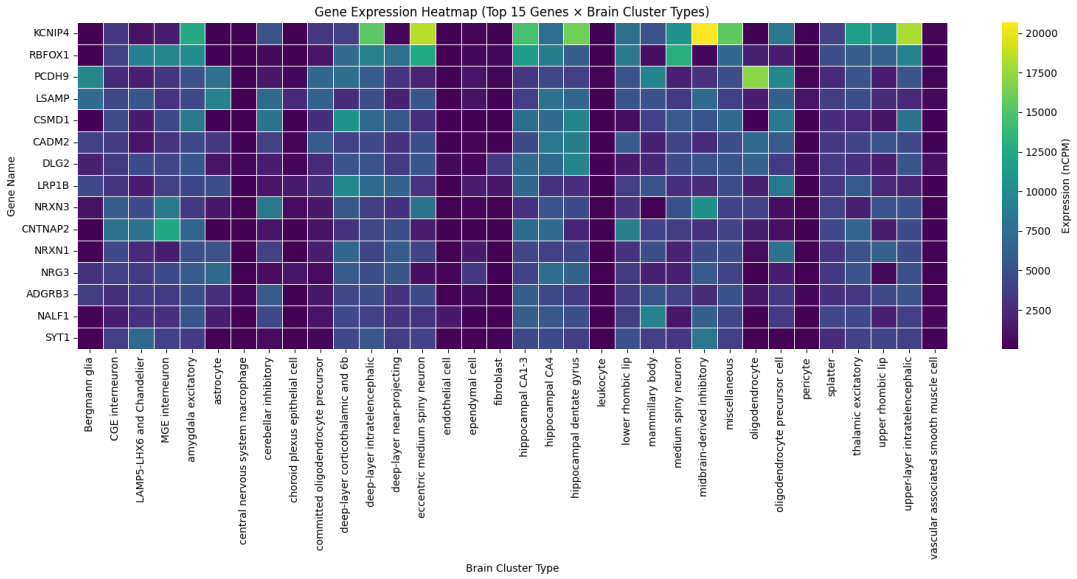
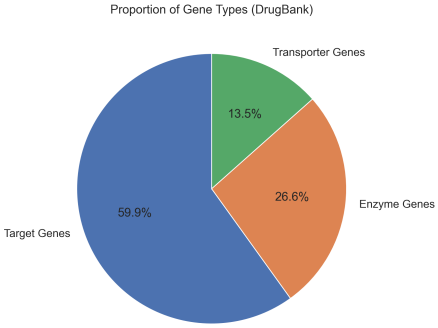
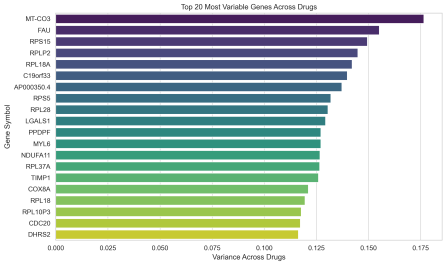


Figure 2: **gene expression patterns across brain cell clusters.**



(a) Proportion of Gene Types



(b) Top 20 Most Variable Genes Across Drugs

Figure 3: Result of Task 2.

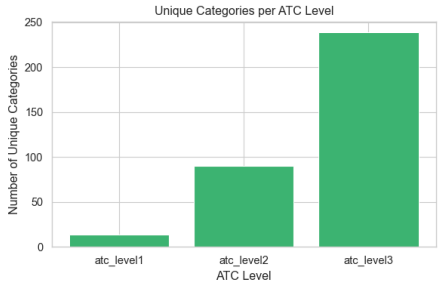
4.4 Gene Signature Integration

Task 4 integrates brain gene expression features and gene signature information into numerical embeddings for each drug. From 2,077 DrugBank drugs with ATC codes, feature extraction was successfully completed for 1,983 drugs (95.5

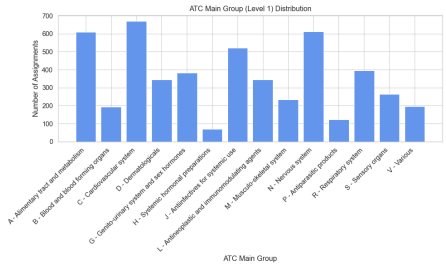
During preprocessing, missing or invalid values were detected and cleaned to ensure data quality. The features were then standardized to approximately zero mean and unit variance, making them suitable for machine learning models.

The feature matrix heatmap (Figure 5a) shows variability across drugs and features, confirming that the embeddings are not constant and that different drugs have distinct numerical profiles. Additionally, the PCA projection of the feature matrix (Figure 5b) demonstrates structured dispersion of samples in the reduced two-dimensional space, indicating that the integrated features capture meaningful variation across drugs.

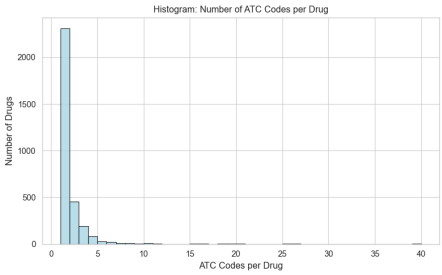
Overall, Task 4 transforms biological information into a structured, standardized, and informative feature representation that supports downstream classification and prediction tasks.



(a) Unique Categories per Level



(b) ATC Level Distribution



(c) Histogram of ATC Codes per Drug

Figure 4: Result of Task 3.

4.5 Drug Category Classification

Task 5 uses a two-stage classification approach.

In Stage 1, four models (Random Forest, Gradient Boosting, SVM, and Logistic Regression) were trained to classify drugs as CNS or Non-CNS. Gradient Boosting achieved the best cross-validation accuracy (0.894)(Figure 6a) . The ROC curve (AUC = 0.98) and confusion matrix (Figure 6c) show strong performance. Although the data is imbalanced in figure 6b (16.5

In Stage 2, a multi-class model was trained to predict 14 ATC categories. Random Forest performed best with a CV accuracy of 0.462. However, training accuracy (0.939) is much higher than CV accuracy, which shows overfitting. Test accuracy (0.451) confirms moderate performance (Figure 6d

Common categories like N and L perform better, while rare categories (H, P, V) perform poorly due to class imbalance. Overall, Stage 1 is reliable, while Stage 2 needs improvement to reduce overfitting and handle imbalance better.

4.6 Prediction and Evaluation.

In Task 6, the trained classification model is applied to unseen test drugs to predict their therapeutic category (ATC Level 1). The prediction (Figure 7) is based on the gene-derived feature embeddings created in Task 4, which integrate brain gene expression patterns and drug target genes. The model does not use category information as input; ATC categories are used only as labels during training.

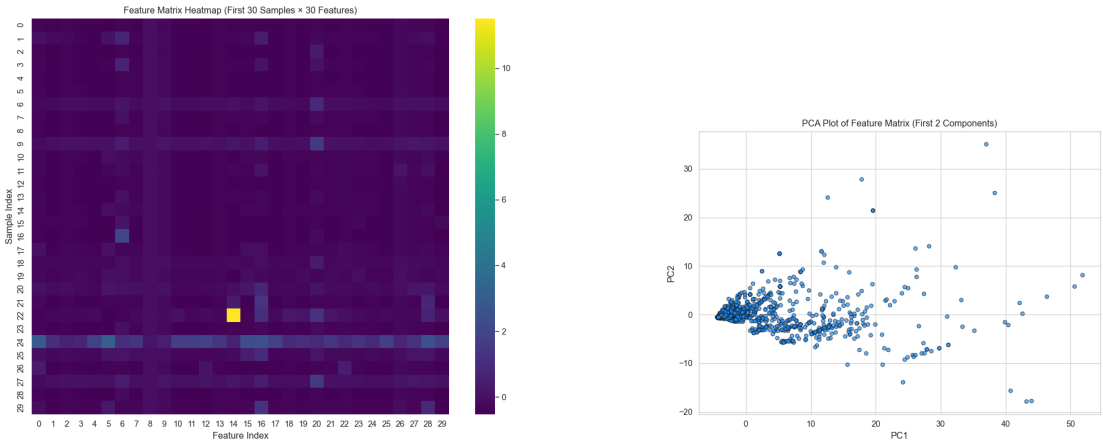
On the test set (397 drugs), the model achieved a Top-1 accuracy of 45.1

These results show that while exact single-label prediction is challenging, the system reliably ranks the correct therapeutic category among its top suggestions, making it suitable for decision-support scenarios. supports using brain data in the prediction model.

5 User Interface and Explainer Agent

A web-based user interface was developed to allow biomedical researchers to use the system without programming knowledge. The interface communicates with the trained Neuro-Symbolic (NeSy) pipeline through a backend API. Users can input the target genes of a candidate drug and receive:

- Top-3 predicted therapeutic categories (ATC Level 1) with probabilities,



(a) The Feature Matrix Heatmap

(b) The PCA Projection of the Feature Matrix

Figure 5: Result of Task 4.

- A CNS relevance score indicating the likelihood of central nervous system involvement,
- Brain Evidence Metrics (BES and BSR),
- Gene coverage information (percentage of input genes found in the database),
- A structured explanation generated by an evidence-based agent.

Prediction probabilities alone are insufficient for biomedical decision-making. Therefore, the system provides biological evidence metrics to assess whether the predicted category is supported by brain gene expression data.

5.1 Brain Evidence Metrics

To quantify brain-related support, two complementary metrics are defined.

Brain Evidence Strength (BES)

[KMRR17]

BES measures the overall expression level of the input genes across brain cell cluster types:

$$\text{BES} = \text{median}\left(E_{g,r}\right)$$

where $E_{g,r}$ represents the normalized expression value of gene g in brain cell cluster type r.

The median is used instead of the mean to reduce the influence of extreme outliers and highly expressed genes, making the metric more robust.

Brain Specificity Ratio (BSR)

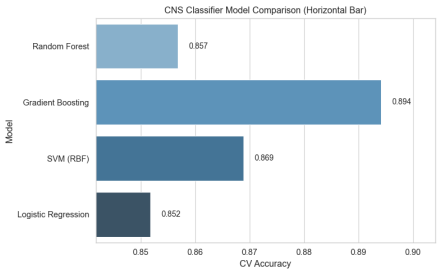
BSR measures how specific the gene expression is to brain tissue compared to non-brain tissues:

$$\text{BSR} = \frac{\max\left(E_{\text{brain}}\right)}{\max\left(E_{\text{body}}\right)}$$

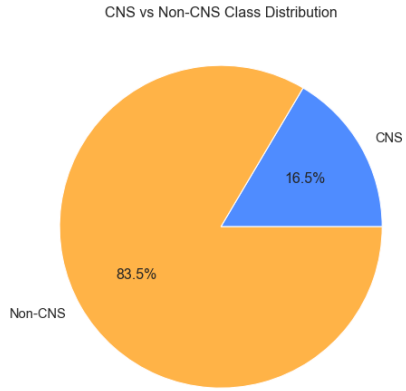
where:

- $\max(E_{\text{brain}})$ is the maximum expression across brain cell cluster types,
- $\max(E_{\text{body}})$ is the maximum expression across non-brain tissues.

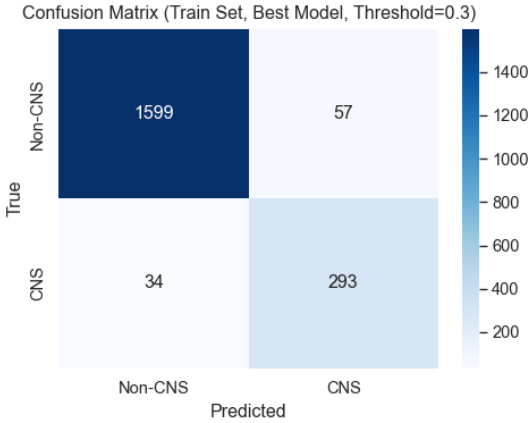
A value BSR > 1 indicates brain-enriched expression, while lower values suggest distributed or peripheral expression.



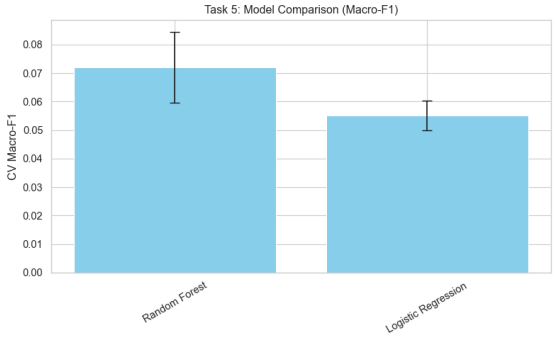
(a) Stage1:CNS/Non-CNS classification model comparison



(b) CNS vs Non-CNS Class Distribution



(c) CNS vs Non-CNS Confusion Matrix



(d) Model macro-F1 Comparison for ATC Classification

Figure 6: Result of Task 5.

Brain Sufficiency Flag (BSF)

Brain evidence is considered sufficient when both strength and specificity exceed predefined thresholds:

$$\mathrm{BSF} = (\mathrm{BES} \geq \tau_{\mathrm{strength}}) \ \wedge \ (\mathrm{BSR} \geq \tau_{\mathrm{ratio}})$$

where default thresholds are:

$$\tau_{\mathrm{strength}} = 0.3, \qquad \tau_{\mathrm{ratio}} = 0.6.$$

5.2 Rationale for the Formulation

The BES formulation captures the overall magnitude of brain expression while remaining robust to outliers through the use of the median. The BSR formulation reflects relative tissue specificity, a common principle in transcriptomic analysis where enrichment ratios are used to determine tissue-selective expression.

Together, BES measures *expression strength*, and BSR measures *expression specificity*. Their combination provides a biologically interpretable assessment of whether a prediction is sufficiently supported by brain-related molecular evidence.

The explainer agent integrates these metrics with model probabilities and CNS scoring to provide transparent, evidence-based reasoning rather than purely black-box predictions.

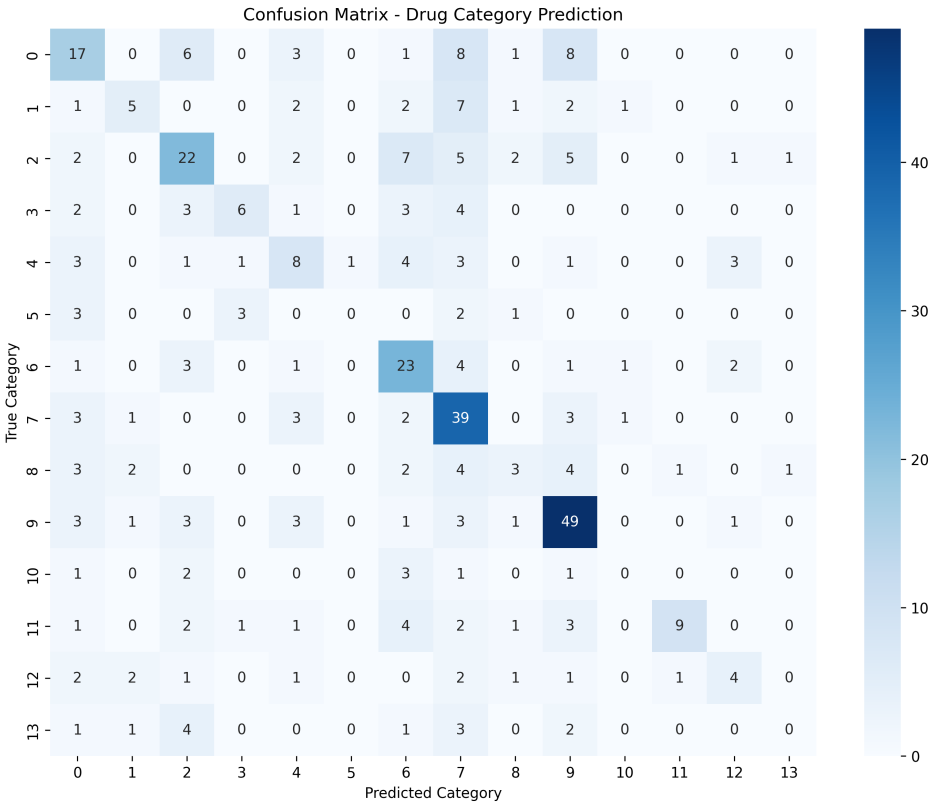


Figure 7: **Task6: Test Confusion Matrix**

6 Conclusion, Limitations, and Future Work

This project presents a neuro-symbolic framework for predicting drug therapeutic categories based on target genes and brain cell-type-specific expression data. The system integrates DrugBank targets, gene expression signatures, and brain transcriptomic data to generate biologically informed feature embeddings for classification.

Results show strong performance for CNS vs Non-CNS classification (about 87–89

However, the multi-class model shows clear overfitting, as training accuracy is much higher than validation accuracy. This is mainly due to limited data size and class imbalance. Because the project was developed and executed locally, lightweight models were used. With more computational resources and larger datasets, deeper models such as neural networks could be applied and may provide better generalization.

Another limitation is that the system relies mainly on gene expression distribution. Real biomedical prediction problems involve many additional factors, such as blood biomarkers, pharmacokinetics, protein interactions, and clinical data. Incorporating these features could significantly improve performance.

Overall, this framework can be considered a first step toward building an interpretable AI-based prediction tool for biomedical research. While promising, further model optimization, data expansion, and integration of additional biological factors are necessary for real-world deployment.

How AI Assisted This Project.

AI tools were used to support research, coding, and writing in this project.

Theoretical research: ChatGPT (GPT-5.2) was used to explore related concepts and find relevant papers. When using AI for scientific information, I requested sources and reviewed them independently to reduce hallucination and incorrect data.

System design and coding: The neuro-symbolic system was designed using the Tool4Boxology

framework from TIB Institute. The structured design helped generate initial code using GitHub Copilot in VS Code. All code was manually reviewed, corrected, tested, and improved by me.

Writing and interpretation: ChatGPT was used to improve clarity and rephrase explanations. All results and numerical values were verified directly from model outputs before being included in the report.

AI tools were used as supportive assistants, while all final decisions, validation, and interpretations were completed independently.

Data and Code Availability

The datasets used in this project are publicly accessible through Google Drive:

https://drive.google.com/drive/folders/1rnPldk5j06xEjuf3Tqk_dHZKW6ji0HOq?usp=sharing

The complete source code for the Neuro-Symbolic drug category prediction system is available in the following GitHub repository:

https://github.com/Mahsa-for/DrugCategoryPrediction

All scripts, trained models, and intermediate outputs required to reproduce the results are included in the repository.

References

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